

Manufacturing Quality Intelligence System (MQIS): Prototype-Stage Progress Report

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Prototype-Stage Report for [UCS503]

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1. Purpose of This Report

This report documents the current state of the Manufacturing Quality Intelligence System (MQIS) prototype, six weeks into the project timeline laid out in our proposal. It records what has been implemented and verified working end-to-end, what is partially working and still being refined, and what remains planned for the next phase. This is a checkpoint, not a final deliverable — some components described below are functional but not yet tuned for accuracy or robustness.

2. Recap: What MQIS Is

MQIS is a defect-detection and quality-reporting pipeline for a manufacturing line. It compares captured part images against a known-good reference to detect and classify defects, logs every inspection to a database, computes live rejection analytics, and uses a locally-hosted LLM with retrieval-augmented generation (RAG) to answer questions and generate shift reports grounded in factory knowledge documents.

3. Current Progress Summary

3.1 Working End-to-End

The following pipeline is implemented and runs successfully as a connected flow, not just as isolated modules:

- An inspection can be triggered for a selected production line.
- The vision module compares the captured image against a reference image using grayscale differencing, thresholding, and contour extraction, and returns a defect classification with a bounding box, severity, and confidence score.
- Every inspection result (pass or reject) is written to the database with a timestamp, line ID, and defect metadata.
- A knowledge base of factory notes has been embedded into a local vector store, and retrieval against that store returns relevant chunks for a given query.

This satisfies the Milestone 1 target from our original proposal: working detection, a logging pipeline, and functioning retrieval, all integrated rather than developed in isolation.

3.2 Partially Working / In Progress

- Detection accuracy: the current thresholding parameters were tuned against an initial sample set and still produce an elevated false-positive rate on certain lighting conditions. Further calibration is planned (see Section 6).
- Live analytics dashboard: per-line rejection rates and status flags are implemented but not yet fully integrated into the main dashboard view.
- RAG report generation: the retrieval step works correctly, but full natural-language shift-report generation (combining retrieved knowledge with live summary data into a coherent report) is still being wired into the pipeline end-to-end.

3.3 Not Yet Started

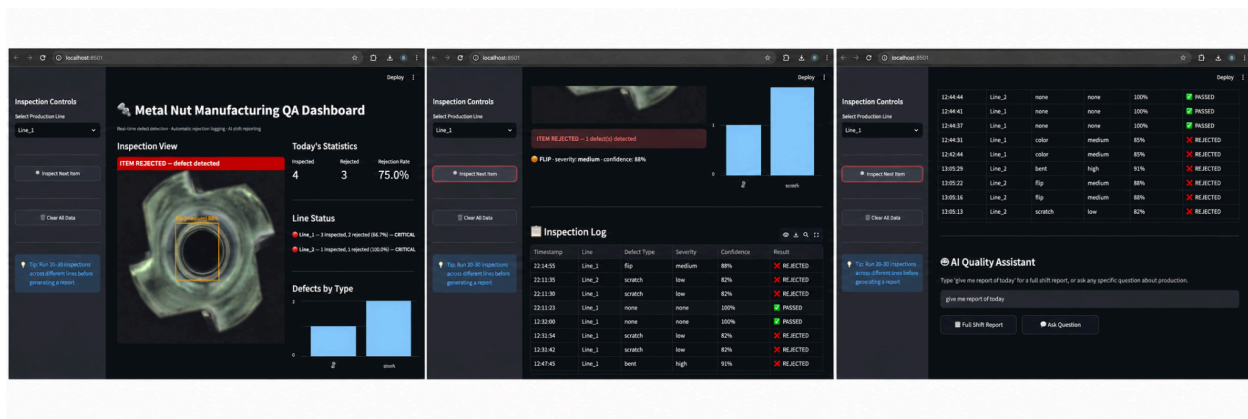
- Free-text question-answering interface for operators.
- Full system integration testing across all modules simultaneously.
- Final accuracy validation against a held-out test set.

4. System Architecture (As Built So Far)

The architecture matches the 3-tier design from our proposal:

- Frontend: a Streamlit dashboard for triggering inspections and viewing results.
- Backend/logic: independent Python modules for detection, database logging, and RAG retrieval, integrated into a single application entry point.
- Data layer: SQLite for event logging; a local persistent vector store for the factory knowledge base.

5. Current Prototype — Screenshot



- Accuracy improvements: re-tune differencing thresholds and contour area cutoffs against a larger labelled sample set to reduce false positives; evaluate whether an adaptive threshold (rather than a fixed value) is needed across varying lighting.
- Full analytics integration: connect the existing rejection-rate and status-flag computations directly into the live dashboard view.
- Complete RAG reporting: finish wiring retrieved knowledge and live summary data into full shift-report generation, and add the free-text question-answering interface.
- System-wide integration testing: run a fixed batch of inspections across all three simulated lines and validate rejection rates and reports against manually computed ground truth, as specified in our proposal's evaluation plan.

7. Status Against Original Timeline

Per our proposal's 15-week plan, Weeks 1–6 targeted working detection, an integrated logging pipeline, and functioning retrieval as Milestone 1 — all three are in place. Weeks 7–11 (threshold tuning, full analytics, full RAG reporting, dashboard integration) are the immediate next steps and are already underway, as described in Section 3.2 above.

8. Summary

The MQIS prototype currently demonstrates a working, integrated pipeline from image capture through defect detection, database logging, and knowledge retrieval. Accuracy tuning, full analytics integration, and complete RAG-based reporting are in active development and are on track with our original 15-week timeline. The next progress checkpoint will report on threshold tuning results, the fully integrated analytics dashboard, and end-to-end shift report generation.